Mykola Zhuk

m.zhuk.ca@gmail.com (437) 259-5516 Toronto, Ontario LinkedIn GitHub Portfolio

Experience

Software Developer - Contract Jun 2026 – Aug 2026 | Remote
B2B SaaS Startup
- Built a **Python pipeline** extracting 20 corporate-event types from 2,936 articles into 1,332 records, with **99.7% passing client-defined schema validation**.
- **Deployed and tested on Azure** using Container Apps Jobs, Service Bus, and Blob Storage, with PostgreSQL-backed idempotency, automated tests, and audit logs.
- Reduced **LLM calls by 17.6% and token usage by 17.9%** through caching and selective inference.
- Estimated operating costs of **US$400/month for 30,000+ articles**, versus the founder's US$6,000/month estimate for existing infrastructure and external services.

Backend Engineer - Volunteer Oct 2025 – Mar 2026 | Remote
Tacmedbat
- Built NestJS/MongoDB APIs and a custom admin interface enabling the customer to edit Ukrainian/English landing-page content. Designed a public endpoint combining **8 collections to support 10+ frontend blocks**.
- Delivered **15+ backend features and fixes** across Cloudinary image handling, **role-based access control**, request validation, payments, and SMTP delivery.

Software Analyst Jun 2024 – Aug 2024 | Toronto
Accenture
- Developed Python/OpenAI API prototypes for retail Q&A and semantic search, supported by data pipelines ingesting and enriching **50,000+ product records**.
- Delivered features with a cross-functional team through Agile sprints, code reviews, and stakeholder demos.

Projects

Full-Stack Engineer Oct 2026 | Toronto
Talk to a Document
- Built and deployed a TypeScript/Next.js application for voice and text conversations with PDFs and YouTube videos, integrating **11 API endpoints**, PostgreSQL, and OpenAI Realtime streaming over WebRTC.
- Implemented **session recovery and voice/text handoffs** supporting restoration of up to **128 dialogue turns**, validated with **1,660 unit tests and 160 browser mode-switch checks**. **Live Demo**

Full-Stack Engineer Oct 2024 – Dec 2024 | Toronto
Scriptorium
- Built **20+ REST API endpoints** and React/Next.js flows for code forking, search, and execution, with JWT authentication, role-based access, and GitHub/Google OAuth 2.0 sign-in.
- Built **Docker execution environments for 10+ languages** with execution-time and memory limits; deployed the application on **AWS EC2**. **Live Demo (AWS)**

Backend Engineer Oct 2024 – Dec 2024 | Toronto
ProjectOllon (Industry-Partnered MVP)
- Designed a Prisma schema for **100+ files and 2,000+ code components**, reducing full-scan time by **80%** in project evaluation.
- Built **static-analysis checks for style and duplication** and integrated OpenAI/Cohere models to flag code complexity; helped deliver the code-review MVP to industry partner Ollon.

Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Bash

Frameworks & Data: React, Next.js, Node.js, NestJS, PostgreSQL, MongoDB, Prisma, PyTorch, scikit-learn

Cloud & Testing: Azure, AWS EC2, Docker, GitHub Actions, pytest, Jest, Postman, Git, Linux, Playwright

Education

Honours B.Sc, Mathematics Jun 2025
University of Toronto
Relevant Courses: Software Design, Machine Learning, Data Structures and Algorithms, Web Programming, System Programming, Computer Organization